

MAT 310, Handout 4: Conditional Probability, Independence, Probability Laws (2.7, 2.8)

Conditional Probability The probability of A given B is denoted by:

$$P(A|B) =$$

Example 1. Roll 2 dice. Let $A = \{\text{roll 2 odd numbers}\}$ and $B = \{\text{at least one roll is odd}\}$. Find $P(A|B)$.

Sometimes the knowledge of whether an event B has occurred may affect the probability that a different event A will occur.

Example 2. NFL: Bills vs. Rams. Is the probability that the Bills win dependent upon:

- whether Josh Allen is injured?
- the revenue earned from commercials during the game?
- which team gets penalized the most?

Note: If A is dependent on B , then $P(A|B) \neq P(A)$.

Definition: A and B are **independent** events *if and only if* any of the following are true:

1. $P(A|B) =$
2. $P(B|A) =$
3. $P(A \cap B) =$

Note: These three statements are mathematically equivalent. If any of them are not true, then none of them are true, and we say that A and B are **dependent**.

Events $A_1, A_2, \dots, A_n$ are **mutually independent** if and only if, for all subsets $I \subseteq \{1, 2, \dots, n\}$,

$$P(\cap_{i \in I} A_i) = \prod_{i \in I} P(A_i).$$

Example 3. Flip 2 coins. Let A = flip exactly one Head; B = 1st flip was a Head; C = flip at least one Head. Determine the following:

- Are A and B independent?
- Are A and C independent?
- Are B and C independent?

Multiplicative Law: $P(A \cap B) =$

Similarly, $P(A \cap B) =$

Proof:

Note: if A and B are independent, then $P(A \cap B) =$

Note: the Multiplicative Law can be extended to any number of events.

- $P(A \cap B \cap C) =$

- $P(A_1 \cap A_2 \cap \dots \cap A_k) =$

Additive Law: $P(A \cup B) =$

Proof:

Note: if A and B are disjoint, then $P(A \cup B) =$

Example 4. Write out the Additive Law for 3 events.

Complementary Law: $P(A) + P(\overline{A}) =$

Proof:

Example 5. Roll 2 dice. Find the probability that you roll at least one even number.

Example 6. Find the probability that you are dealt at least two cards that are Jack or better (i.e., Jack, Queen, King, Ace) in a 5-card hand.

Example 7. Suppose there are 20 people in a room. Find the probability that at least two of them share the same birthday.